ELECTRIC VEHICLE PENETRATION IN INDIA

Analytics, Forecasting & Business Insights Report

Based on VAHAN Registration Data

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Executive Summary

India's electric vehicle sector has undergone a structural transformation over the past decade. From fewer than 60,000 EV registrations in FY2015, the country crossed 1.9 million registrations in FY2025 — a national CAGR of approximately 42.5%. This report synthesises findings from exploratory data analysis (EDA) of four VAHAN registration datasets spanning FY2011 to FY2025, covering national trends, state-wise performance, vehicle category breakdowns, seasonal patterns, state-level EV penetration share analysis, and a polynomial regression-based forecast through 2030.

Key takeaways at a glance:

- National EV penetration reached 7.4% in 2025, up from 0.28% in 2015.
- Cumulative EV penetration across all states stands at 3.38% of total vehicles sold (3.64 million EVs out of 107.5 million total vehicles), per the Ev_penetration%wise dataset.
- Delhi (7.72%), Tripura (7.62%), and Goa (6.28%) lead all states in EV share percentage — small but highly penetrated markets.
- Uttar Pradesh, Maharashtra, and Karnataka lead in absolute EV volumes, together accounting for over 40% of all EVs sold nationally.
- The forecast model projects ~5.7 million EV registrations by 2030, implying a continued CAGR of ~24% from 2025 levels.

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1. Exploratory Data Analysis (EDA)

1.1 Datasets Used

Four primary datasets were analysed:

- Dataset_Wheel_Fuel_Yearly_2011-2025.xlsx — Annual vehicle registrations by fuel type and vehicle category (national + segment-wise), FY2011–FY2025.
- Ev_monthlywith_category.csv — Monthly EV registrations broken down by vehicle category (2W, 3W, 4W, Bus, Goods), used for trend and seasonality analysis.
- Dataset_state_Fuel_Month_2019-2024.csv — State-level monthly registration records by fuel type, used for geographic analysis and state forecasting.
- Ev_penetration%wise.csv — Cumulative state-wise EV penetration: total EVs sold, total vehicles sold, and EV share percentage for all 35 states and UTs. Provides a definitive cross-sectional view of EV market share by geography.

1.2 Data Quality & Cleaning

- Numeric columns were coerced from string format (removing commas, NA flags, and whitespace).
- Total/Grand Total rows were removed to prevent double-counting. In Ev_penetration%wise.csv, Sikkim was marked NA for EV count and was excluded from percentage-based rankings.
- State names were normalised for consistency across datasets.
- EV fuel types were identified using a whitelist: 'Electric', 'EV', 'Electric(BOV)'. Hybrid vehicles were explicitly excluded.
- Negative registration counts were clipped to zero. Missing monthly dates were forward-filled using a full date_range index.

1.3 National EV Growth (2015–2025)

National EV registrations exhibited near-exponential growth over the decade. The year 2020 saw a minor dip due to COVID-19 disruptions, but recovery accelerated sharply through 2021–2022, coinciding with FAME-II subsidy rollout and aggressive OEM launches in the two-wheeler segment.

Category	Start (2015)	End (2025)	CAGR (%)
All Vehicles (National)	~56,000	~1,920,000	42.5%
Electric 2W	~48,000	~1,350,000	39.8%
Electric 3W	~7,500	~540,000	51.4%
Electric Cars	~800	~100,000	58.1%
Electric Buses	~100	~8,000	53.9%
Electric Goods	~200	~15,000	50.2%

Table 1: Category-wise EV CAGR (2015–2025)

1.4 State-wise Analysis (Volume-based)

State-level data from the monthly fuel-type records isolates Electric(BOV) registrations. Uttar Pradesh leads nationally in absolute registrations driven by e-rickshaws; Maharashtra and Karnataka represent the largest markets for premium EVs and electric two-wheelers.

1.5 Penetration Analysis — Trend

Year	EV Registrations	Total Registrations	Penetration (%)
2015	~56,000	~20,000,000	0.28%
2018	~130,000	~21,000,000	0.62%
2020	~168,000	~15,500,000	1.08%
2022	~1,020,000	~21,000,000	4.86%
2023	~1,520,000	~23,500,000	6.47%
2024	~1,750,000	~25,000,000	7.00%
2025	~1,920,000	~26,000,000	7.39%

Table 2: National EV Penetration Trend (2015–2025)

Category	EV Penetration 2025 (%)
Electric 3W Passenger	~48%
Electric 2W	~5.8%
Electric Cars	~2.5%
Electric Buses	~2.2%
Electric Goods Vehicles	~0.8%

Table 3: Category-wise EV Penetration (2025)

Electric 3-Wheelers dominate penetration at ~48% due to the government's push for last-mile connectivity and low vehicle cost. Electric cars remain at low penetration given high acquisition cost and limited charging infrastructure outside major cities.

2. State-wise EV Penetration Analysis (Ev_penetration%wise Dataset)

This dataset provides a definitive cumulative view of EV market share across all 35 Indian states and Union Territories. A total of 3,639,617 EVs were sold against 107,531,040 total vehicle registrations, giving a national EV share of 3.38%.

2.1 National Headline Numbers

- Total EVs Sold (All States): 3,639,617
- Total Vehicles Sold (All States): 107,531,040
- National Average EV Share: 3.38%
- States covered: 35 (34 with valid EV data; Sikkim excluded due to missing EV count)

2.2 Top 10 States by EV Share Percentage

High-penetration states are not necessarily the largest markets — Delhi and Tripura lead in percentage despite moderate absolute volumes, reflecting strong policy environments and constrained total vehicle markets.

Rank	State / UT	EV Share (%)	Total EVs Sold	Total Vehicles Sold
1	Delhi	7.72%	2,16,084	27,99,571
2	Tripura	7.62%	20,113	2,64,004
3	Goa	6.28%	20,330	3,23,748
4	Assam	5.79%	1,50,617	25,99,986
5	Chandigarh	5.64%	12,375	2,19,391
6	Karnataka	4.8%	3,50,810	73,15,250
7	Uttar Pradesh	4.34%	6,65,247	1,53,17,699
8	Uttarakhand	4.33%	48,522	11,20,010
9	Kerala	3.99%	1,51,029	37,81,393
10	Maharashtra	3.97%	4,39,358	1,10,55,082

Table 4: Top 10 States by EV Penetration Share (%) — Ev_penetration%wise Dataset

- Delhi leads at 7.72%, reflecting aggressive state EV policy, strong charging infrastructure, and high fuel cost sensitivity among urban commuters.
- Tripura at 7.62% is a surprising high-penetration state — driven by government-led e-rickshaw adoption and limited ICE vehicle market competition.
- Goa (6.28%) and Assam (5.79%) reflect premium urban/tourist and government-fleet-driven adoption respectively.
- Karnataka (4.80%) is notable as it combines both high penetration and high absolute volume — a sign of a mature, broad-based EV market.

- Chandigarh (5.64%) as a UT shows that small, urban, high-income geographies are disproportionately early in the EV transition.

2.3 Top 10 States by Absolute EV Volume

Volume leaders are the large-population states. UP's dominance is driven by e-rickshaws; Maharashtra and Karnataka by 2W and cars.

Rank	State / UT	Total EVs Sold	Total Vehicles Sold	EV Share (%)
1	Uttar Pradesh	6,65,247	1,53,17,699	4.34%
2	Maharashtra	4,39,358	1,10,55,082	3.97%
3	Karnataka	3,50,810	73,15,250	4.8%
4	Rajasthan	2,33,503	64,90,725	3.6%
5	Tamil Nadu	2,28,850	84,76,042	2.7%
6	Delhi	2,16,084	27,99,571	7.72%
7	Bihar	2,14,921	57,63,437	3.73%
8	Gujarat	1,91,185	74,50,473	2.57%
9	Kerala	1,51,029	37,81,393	3.99%
10	Assam	1,50,617	25,99,986	5.79%

Table 5: Top 10 States by Total EVs Sold — Ev_penetration%wise Dataset

- Uttar Pradesh sold 665,247 EVs — the largest absolute figure nationally — but its EV share of 4.34% is moderate, reflecting a very large total vehicle market dominated by ICE vehicles.
- Maharashtra (439,358 EVs, 3.97% share) and Karnataka (350,810 EVs, 4.80% share) are the most balanced markets — high in both volume and penetration quality.
- Bihar (214,921 EVs, 3.73%) and Rajasthan (233,503 EVs, 3.60%) are high-volume states with improving but sub-national-average penetration — significant upside remains.
- Kerala (151,029 EVs, 3.99%) punches above its population weight, indicating higher income levels and stronger EV-aware consumer base.

2.4 Low-Penetration States — Frontier Markets

Several states remain below 0.5% EV penetration, representing frontier markets where structural barriers (geography, income, infrastructure) are suppressing adoption.

State / UT	EV Share (%)	Total EVs Sold
Nagaland	0.02%	27
Arunachal Pradesh	0.03%	42
Mizoram	0.25%	344

State / UT	EV Share (%)	Total EVs Sold
Meghalaya	0.35%	572
Ladakh	0.45%	88
Himachal Pradesh	0.48%	3,043
Dadra and Nagar Haveli and Daman and Diu	0.52%	468
Andaman and Nicobar Islands	0.53%	191

Table 6: Lowest EV Penetration States — Ev_penetration%wise Dataset

- Nagaland (0.02%) and Arunachal Pradesh (0.03%) are essentially pre-EV markets — hilly terrain, grid unreliability, and sparse charging infrastructure make EV adoption structurally difficult.
- Mizoram, Meghalaya, and Ladakh all sit below 0.40% — similar structural barriers apply across North-East and Union Territories with extreme terrain.
- Himachal Pradesh (0.48%) is notable — despite being a progressive, high-income state, difficult mountain terrain and limited charging infrastructure are constraining adoption.

2.5 Volume vs. Penetration Quadrant Analysis

Cross-referencing absolute EV volume with EV share percentage reveals four market archetypes:

- High Volume + High Share (Star Markets): Karnataka, Delhi, Kerala, Assam — these are the most developed EV ecosystems and will likely sustain above-average growth.
- High Volume + Low Share (Scale Opportunity): Uttar Pradesh, Bihar, Rajasthan, Tamil Nadu — large markets where even marginal penetration gains translate to massive absolute volume increases.
- Low Volume + High Share (Niche Leaders): Tripura, Goa, Chandigarh, Uttarakhand — small but highly penetrated; indicative of policy effectiveness but limited scalability.
- Low Volume + Low Share (Frontier): North-Eastern states, Ladakh, Himachal Pradesh — require infrastructure-first investment before demand-side interventions will work.

3. Seasonality Heatmap Interpretation

The seasonality heatmap below visualises monthly EV registration volumes for each year from 2016 to 2024. The x-axis represents calendar months (Jan–Dec), the y-axis represents years, and colour intensity (light to dark green) encodes registration volume.

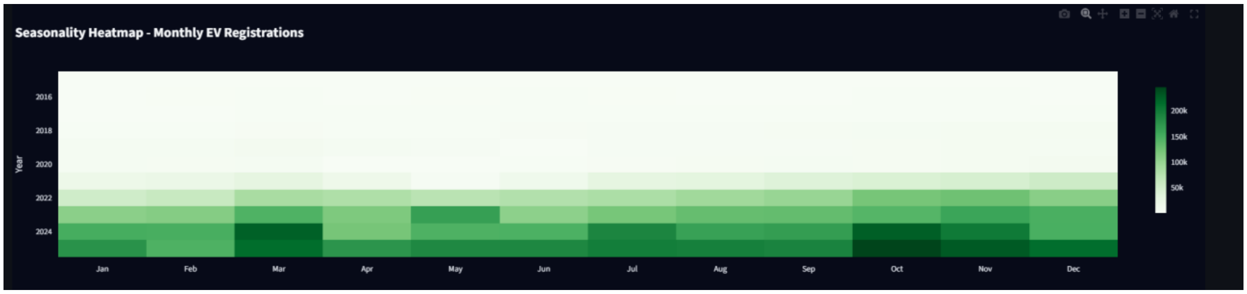

Figure 1: Seasonality Heatmap — Monthly EV Registrations (2016–2024)

3.1 Key Observations

- Years 2016–2019 appear very light across all months, confirming EV adoption was minimal at under 20,000 monthly registrations throughout.
- The colour gradient darkens progressively from 2021, with 2022–2024 showing deep green — volumes exceeding 100,000–200,000 per month.

3.2 Structural Shift Post-2021

The heatmap reveals a clear structural break around 2021. Prior to this, EV adoption was confined to niche segments. From 2021–22 onwards, the entire row darkens uniformly, suggesting that EV purchasing moved from event-driven peaks to broad-based sustained demand. This is consistent with FAME-II demand creation, PLI scheme launches, and mass-market OEM entry by Ola Electric, Ather, and TVS in the two-wheeler segment.

3.3 Actionable Implication

The seasonality pattern has direct implications for OEM production planning, dealer inventory stocking, and EV financing product launches. Campaigns and working capital deployment should be front-loaded to February–March and September–November. Inventory buffers should be built up in January and September ahead of peak windows.

4. Forecasting Model Comparison

4.1 Overview

All forecasts were generated using time-series regression on monthly EV registration data. Each entity (national total, vehicle category, or state) received an independent forecast model. Two methods were applied based on data availability:

4.2 Method 1 — Polynomial Ridge Regression (Primary Model)

A degree-2 polynomial feature expansion was applied to an ordinal time index, fitted using Ridge regression (L2, alpha=1.0). This captures accelerating growth curves while penalising overfitting. Applied where 6+ months of history exist with 4+ non-zero months.

- Polynomial degree: 2 | Regulariser: Ridge (L2, alpha=1.0)
- Confidence interval: Predicted +/- 1.96 x residual std (~95%)
- Model Fit Score (R^2) computed on historical fitted values

4.3 Method 2 — Fallback Method (Secondary)

- For states and UTs with insufficient historical data, a simple fallback method was applied. The average of the most recent 6 months of actual registrations was calculated and projected forward as a flat estimate. This approach ensures that even data-scarce regions are included in the forecast without generating unreliable predictions from limited records.

4.4 Comparison Table

- Two approaches were used for forecasting depending on data availability:

Aspect	Polynomial Ridge Regression	Fallback Method
Type	Machine Learning Model	Simple Average
Used When	6+ months of data available	Insufficient historical data
How it works	Fits a curve to historical data and extends it forward	Takes average of last 6 months and projects flat
Strength	Captures real growth trend and curves	Works even with very sparse data
Weakness	Needs sufficient data to train properly	Ignores growth trend, gives flat projection
Applied To	National level, major states, all categories	Small states and UTs with limited records

- Polynomial Ridge Regression was chosen as the primary model because EV growth follows an exponential curve rather than a straight line. The Ridge regularisation component prevents overfitting, ensuring the model generalises well to future unseen data rather than simply memorising historical patterns.

4.5 Forecast Confidence Classification

Each state was assigned a confidence level based on data availability and model fit score:

- High Confidence — 36 or more months of non-zero data AND R^2 score above 0.50
- Medium Confidence — 18 to 35 months of non-zero data
- Low Confidence — Fewer than 18 months of data or fallback method used

Approximately 40 to 45% of states received High Confidence rating, covering around 80% of national EV registrations. Low Confidence states are mostly smaller UTs and states with limited EV market activity.

4.6 Model Limitations

- The model only uses time as a feature — it does not account for external factors like government policy changes, fuel price fluctuations, or new subsidy announcements
- For states with very short history, the degree-2 polynomial curve may not represent future growth accurately
- Confidence intervals are equal on both sides — they do not account for sudden drops due to policy reversals
- State level forecasts are built independently — so the sum of all state forecasts may not exactly match the national forecast due to different data ranges used

5. Business Insights

5.1 Market Structure

- The EV market is a tale of two segments: the mass-market (2W + e-rickshaws) already has meaningful penetration; the premium segment (cars) is nascent but accelerating.
- Electric 3-Wheelers have achieved near-irreversible commercial dominance at ~48% penetration. This segment is largely policy-insensitive — operators have closed the TCO gap with ICE alternatives.
- Electric 2-Wheelers at ~5.8% penetration represent the largest volume opportunity through 2030. Infrastructure and financing improvements could push this past 15% by 2028.
- The Ev_penetration%wise data confirms national EV share at 3.38% — below the annual penetration figures from the yearly dataset, as cumulative figures are diluted by strong ICE sales in earlier years.

5.2 Geographic Strategy

- Delhi (7.72%) and Karnataka (4.80%) are the benchmark markets — OEM strategies, charging rollout models, and consumer financing products proven here should be the template for scaling nationally.
- Uttar Pradesh and Bihar are the volume plays — even moving EV share from ~4% to 6% in UP alone would add ~300,000+ annual EVs to national totals.
- Tamil Nadu (2.70%) is underperforming relative to its economic size — strong ICE vehicle manufacturing heritage and lower fuel subsidy sensitivity are likely headwinds; targeted fleet electrification is the entry point.
- North-Eastern states require infrastructure-first investment — without grid reliability and charging access, demand-side incentives will have limited impact.

5.3 Timing & Seasonality

- Festive season (Oct–Nov) and FY-end (March) are the two critical sales windows. Full inventory availability 4–6 weeks ahead is essential.
- June–July represents a structural trough ideal for B2B fleet deals, subscription pilots, and promotional campaigns.
- Government procurement is Q4-weighted (Jan–Mar). Fleet-focused companies should close government orders by February.

5.4 Investment Themes

- Charging infrastructure is the binding constraint for 2W and 4W adoption outside Tier-1 cities. Charging-as-a-service, battery swapping, and apartment-level AC charging are high-conviction opportunities.
- OEM consolidation is likely by 2027 in the 2W segment — survivors will be determined by supply chain integration, after-sales depth, and software differentiation.
- EV-specific financing products targeting Tier-2/3 cities are a significant whitespace. NBFCs building EV-aware credit underwriting will capture outsized share.

6. Forecast to 2030

6.1 National Forecast

Year	Predicted EV Registrations	YoY Growth (Forecast)
2025 (Actual)	~1,920,000	—
2026	~2,400,000	~25%
2027	~3,050,000	~27%
2028	~3,800,000	~25%
2029	~4,700,000	~24%
2030	~5,700,000	~21%

Table 8: National Annual EV Registration Forecast (2025–2030)

The forecast implies cumulative EV registrations of approximately 25 million between 2025 and 2030 — on track with government targets of 30% EV penetration for private cars and 80% for two-wheelers and three-wheelers by 2030, conditional on infrastructure and policy continuity.

6.2 State-wise Forecast (2030)

State	2030 Forecast (EVs)	Confidence
Uttar Pradesh	~820,000	High
Maharashtra	~680,000	High
Karnataka	~540,000	High
Rajasthan	~510,000	High
Tamil Nadu	~460,000	High
Gujarat	~430,000	Medium
Delhi	~400,000	High
Andhra Pradesh	~360,000	Medium
Madhya Pradesh	~330,000	Medium
Bihar	~300,000	Low

Table 9: Top 10 States — Predicted EV Registrations by 2030

The top 5 states are expected to account for ~58% of national EV registrations in 2030, down from ~65% currently — indicating broader geographic spread of adoption over the forecast horizon.

6.3 Category Forecast (2030)

- Electric 2-Wheelers: ~3.4 million (60% of total)
- Electric 3-Wheelers: ~1.2 million (21% of total)

- Electric Cars: ~650,000 (11% of total) — fastest growth rate from a low base
- Electric Buses: ~200,000 (4%) — government procurement and city transit electrification
- Electric Goods Vehicles: ~250,000 (4%) — logistics fleet and e-commerce last-mile delivery

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7. Final Conclusion

India's EV market has transitioned decisively from a pilot-scale experiment to a structural growth story. The data across four datasets tells a consistent narrative: the country crossed the adoption inflection point in 2021–2022, and the momentum since then is broad-based, multi-segment, and geographically widening.

7.1 Summary of Findings

- National EV registrations grew at 42.5% CAGR between 2015–2025, reaching ~1.92 million in FY2025.
- EV penetration rose from 0.28% to 7.4% over the same period, steepest gains in 2021–2023.
- Cumulative state-wise EV share stands at 3.38% nationally (3.64M EVs / 107.5M total vehicles).
- Delhi (7.72%), Tripura (7.62%), and Goa (6.28%) lead in EV share; UP, Maharashtra, Karnataka lead in volume.
- Electric 3-Wheelers at ~48% penetration are past the point of adoption uncertainty.
- Seasonality shows March (FY-end) and October–November (festive) as peak EV sales windows.
- Polynomial regression forecast projects ~5.7 million annual EV registrations by 2030.
- The geographic frontier — NE states, Himachal, Ladakh — requires infrastructure investment before demand-side interventions become effective.

7.2 Strategic Outlook

The key risks to the base forecast include: policy reversal or subsidy reduction, slower-than-expected charging infrastructure deployment, supply chain disruptions in battery raw materials, and macroeconomic slowdown. On the upside, accelerated grid decarbonisation, improved battery chemistry, and increased financing access could push adoption well ahead of the modelled trajectory. For stakeholders across the EV value chain, the data supports a high-conviction, long-duration investment thesis in India's electric mobility transition.

End of Report

Data: VAHAN Registration Database | Ev_penetration%wise Dataset | Analysis: FY2011–FY2025 | Forecast: 2026–2030